

Null Responses and Positive Energy in Motion–Staticity Geometry: Hermitian Cones, Sen–Witten Lifts, and Quantum Constraints

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Abstract

A translation charge $P = (E, p)$ defines a null response $F_P(n) = E - p \cdot n$ on the projective sphere. In motion–staticity (MS) variables, positivity of all such responses is equivalent to positivity of the Hermitian form $P = E\mathbf{1} - p \cdot \sigma$. We develop the relation between this form, massive observer measurements and gravitational positive energy. The standard null-to-timelike Poisson transform reconstructs the observer energy; its tensor Hessian equation characterizes the four translation modes. A boundary response variance recovers the invariant mass, and a trace-free Hessian defect controls the error of a translation approximation in a fixed reference frame. For Einstein initial data, explicit conformal and orthonormal-frame identities are followed by a matrix polarization of the established Sen–Witten construction. The resulting Gram map yields the null response norm, the mass determinant and the Hermitian metric on the boundary spinor line. Symmetric powers recover the corresponding coherent kernels and holomorphic section norms. Standard Bondi balance then gives an exact radiative staticity law, $\dot{q}_B = -\int |\nu - D_B|^2 \mathcal{L} \, d\Omega$, and a finite decomposition into remnant staticity, radiation dispersion and relative motion. The quantum analysis distinguishes directional operator positivity from positivity of the full Hermitian block, proves a staticity bound for a specified noncommuting kinetic shell, and records the constraints imposed by covariance, ordering and local negative energy. The gravitational constraints, the Sen–Witten existence theory and the Bondi flux law remain independent inputs. The results give a common response formulation of these structures without constructing a quantum ADM operator or fixing Newton’s constant.

1 Introduction

An asymptotically isolated system is tested by observers with different velocities. Its energy response is Lorentz-linear in the observer, while its invariant mass is independent of that choice. The MS construction expresses this distinction through a compact motion variable D and a staticity variable S , with $|D|^2 + S^2 = 1$. The interior $S > 0$ parametrizes massive observers; the boundary $S = 0$ records future null rays. The same projective sphere also supports the spinor lines used in helicity geometry.

The present problem is to relate that kinematical boundary to the positivity of gravitational translation charges. The companion boundary-charge construction identifies the MS-completed charge with ADM momentum after the Einstein normalization is supplied [1–3]. The companion spin analysis studies the rest-state obstruction, Wigner transport and little-group contraction on the compact motion base [4]. Here the central object is the Hermitian response form of a total translation charge. The positional sphere at spatial infinity, the sphere of projective probes

and the sphere of actual radiation directions have different roles. A calibrated asymptotic frame relates their angular descriptions; it does not identify the three spaces.

Several ingredients have established origins. The Lorentz–Hermitian correspondence is standard two-spinor geometry [5]. The positive-mass theorem and its spinorial proof belong to Schoen–Yau, Witten and the analytic work of Parker–Taubes [6–8]; orthonormal-frame energy expressions were developed by Nester and others [9–11]. The null-to-timelike kernel used below is Campiglia’s asymptotic-symmetry kernel [12]. Helicity Chern classes, Borel–Weil reconstruction and coherent-state resolutions are likewise prior results [13–15]. Radiation balance is the standard Bondi–Sachs law [16–18]. These results are used with their original scope.

Our calculations concern their compatibility in response variables. A tensor closure condition separates translation responses from higher angular modes. The same mass determinant is recovered from observer-response variance and from the exterior area of two Sen–Witten response channels. The induced line metric connects that factorization to boundary curvature and coherent kernels. A radiative balance formula extends the collective MS variance to the remnant and radiation charges of an interacting gravitating system. The operator analysis identifies a further condition that is hidden by the classical cone picture: positivity on every classical null probe need not imply positivity on joint quantum probes.

Sections 2 and 3 treat the cone and observer reconstruction. Sections 4–6 give the classical identities and the matrix lift. Section 7 develops the boundary line geometry and its symmetric powers. Radiation is treated in Section 8, followed by quantum consistency conditions in Section 9. We use signature $(-, +, +, +)$ and $c = 1$ in the classical sections. In those units $m = \sqrt{E^2 - |p|^2}$ is the invariant energy scale; restoring units gives $m = m_{\text{phys}}c^2$. The quantum intrinsic scale Q has momentum units, so $m = cQ$ there. The observer’s S and the charge’s normalized staticity are distinguished when both occur.

2 The normalized Lorentz–Hermitian cone

Let σ_i be the Pauli matrices. The MS motion ball has the affine realization

$$\rho(D) = \frac{1}{2}(\mathbf{1} + D \cdot \sigma), \quad \det \rho(D) = \frac{S^2}{4}, \quad |D|^2 + S^2 = 1. \quad (1)$$

Its eigenvalues are $(1 \pm |D|)/2$. The closed ball is therefore the trace-one section of the positive Hermitian cone. The interior has rank two; its extreme points have rank one and are

$$\Pi_n = \frac{1}{2}(\mathbf{1} + n \cdot \sigma) = \frac{\zeta \zeta^\dagger}{\zeta^\dagger \zeta}, \quad |n| = 1. \quad (2)$$

The spinor ζ is defined up to a nonzero complex scalar, giving $\mathbb{CP}^1 \simeq S^2$. Rank-one matrices generate extreme rays of the unnormalized cone and are extreme points of its trace-one section. The rest observer is $\rho(0) = \mathbf{1}/2$.

For $D = dn$, $0 \leq d \leq 1$, the decomposition $\rho(dn) = \frac{1}{2}(1 + d)\Pi_n + \frac{1}{2}(1 - d)\Pi_{-n}$ shows directly that the closed motion ball is the convex hull of its projective boundary. This is a statement about normalized probes; it does not impose a spectrum condition on a physical system.

Represent the translation charge by

$$\mathbf{P} = E\mathbf{1} - p \cdot \sigma, \quad F_P(n) = \text{Tr}(\mathbf{P}\Pi_n) = E - p \cdot n. \quad (3)$$

The massive observer $U = (1, D)/S$ measures $\mathcal{E}_P(U) = (E - p \cdot D)/S$. Thus $S\mathcal{E}_P(U) = \text{Tr}(\mathbf{P}\rho(D))$, and the antipodal decomposition expresses every massive response as a positive combination of two null responses.

Proposition 2.1 (Classical cone tests). *The conditions $F_P(n) \geq 0$ for every $n \in S^2$, $\mathbf{P} \geq 0$, and $E \geq |p|$ are equivalent. They imply $\mathcal{E}_P(U) \geq 0$ for every future unit observer.*

Proof. The eigenvalues of $\mathbf{P}$ are $E \pm |p|$. Unit spinors exhaust its rank-one quadratic tests, and the trace pairing of two positive matrices is nonnegative. $\square$

For $\mathbf{P} > 0$, set $m = \sqrt{\det \mathbf{P}}$ and $V = P/m$. Positive determinant-one Hermitian matrices form $SL(2, \mathbb{C})/SU(2) \simeq H^3$; trace normalization gives the compact MS section. The ideal boundary is reached when $|p|/E \rightarrow 1$. This need not mean $m \rightarrow 0$: at fixed nonzero m it is an infinite-rapidity limit with diverging energy. A finite-energy massless shell additionally requires a radial scale prescription. Likewise, a point of $\mathbb{CP}^1$ specifies a null ray, whose stabilizer is larger than the little group of a scale-fixed null momentum [4, 19].

3 Observer reconstruction and closure of null responses

The weight-three hyperbolic Poisson transform, including its translation modes, is established [12], as is the Hermitian realization of Lorentz geometry [5]. We use these structures to connect the Chern measure of an MS response with observer energy, invariant mass, and a precise test of translation-response closure. The distinction between the transform and its four-dimensional translation sector will be essential.

3.1 Calibrated null directions and observer moments

We use the signature $\eta = (-, +, +, +)$ and write $P = (E, p)$ for the Lorentz-polar representative of a completed translation covector. Thus $F_P(n) = -\eta(P, \ell(n)) = E - p \cdot n$, where $\ell(n) = (1, n)$ and $n \in S^2$. The observer hyperboloid is $H^3 = \{U : \eta(U, U) = -1, U^0 > 0\}$, with metric $g_H = \eta|_{TH^3}$. Define

$$A_U(n) = -\eta(U, \ell(n)), \quad k_U(n) = \frac{\ell(n)}{A_U(n)}, \quad z_U(n) = k_U(n) - U, \quad d\mu_U(n) = \frac{d\Omega(n)}{4\pi A_U(n)^2}. \quad (4)$$

Here k_U is a future null vector of unit energy for U , whereas z_U is a unit vector in $T_U H^3$. The measure μ_U has total mass one. For $P = m_P U$ and nonzero helicity h , the curvature formula of Section 7 gives $d\mu_U = -f_{h,P}/(4\pi h)$. This is a geometric probability measure; identifying it with a physical measurement distribution would require a separate statistical assumption.

Proposition 3.1 (Null moments and observer mass). *Put $f_{P,U} = F_P/A_U$ and $\mathcal{E}_P(U) = -\eta(P, U)$. Then*

$$\int z_U d\mu_U = 0, \quad \int z_U^a z_U^b d\mu_U = \frac{1}{3}(\eta^{ab} + U^a U^b), \quad (5)$$

$$\mathcal{E}_P(U) = \int f_{P,U} d\mu_U = \frac{1}{4\pi} \int_{S^2} \frac{F_P}{A_U^3} d\Omega, \quad \nabla_H \mathcal{E}_P(U) = 3 \int z_U f_{P,U} d\mu_U, \quad (6)$$

$$\begin{aligned} -\eta(P, P) &= \mathcal{E}_P(U)^2 - 3 \operatorname{Var}_{\mu_U}(f_{P,U}) \\ &= \mathcal{E}_P(U)^2 - |\nabla_H \mathcal{E}_P(U)|^2. \end{aligned} \quad (7)$$

These identities hold for every real translation representative P ; the squared invariant can therefore have either sign.

Proof. At $U = (1, 0)$, $z_U = (0, n)$ and μ_U is the uniform round measure, whose first and second moments are 0 and $\delta^{ij}/3$. For a proper orthochronous Lorentz transformation, write $\Lambda \ell(n) = \omega(n) \ell(n')$, with $\omega > 0$. Aberration gives

$$A_{\Lambda U}(n') = \frac{A_U(n)}{\omega(n)}, \quad d\Omega(n') = \omega(n)^{-2} d\Omega(n). \quad (8)$$

Consequently $k_{\Lambda U}(n') = \Lambda k_U(n)$ and $\mu_{\Lambda U} = \Lambda_* \mu_U$, proving (5). Decompose $P = \mathcal{E}_P U + p_U$, with $p_U \in T_U H^3$. Then $f_{P,U} = \mathcal{E}_P - \eta(p_U, z_U)$ and $\nabla_H \mathcal{E}_P = -p_U$. The moment identities give its mean, gradient, and variance $|p_U|^2/3$, hence (7). $\square$

The factor A_U^{-1} in $f_{P,U}$ calibrates each null direction to the observer's unit energy. Averaging the uncalibrated F_P with μ_U would not give $\mathcal{E}_P(U)$. In the reference rest frame, writing $\langle \cdot \rangle = (4\pi)^{-1} \int_{S^2} (\cdot) d\Omega$, the reconstruction reads $E = \langle F_P \rangle$, $p = -3\langle n F_P \rangle$, and $m_P^2 = E^2 - 3 \text{Var}(F_P)$. For a future timelike charge, the MS parameters $D = p/E$ and $S = m_P/E$ therefore satisfy $|D|^2 + S^2 = 1$. Angular response variance measures the observer-dependent momentum part of the same invariant mass relation.

3.2 Relative response intervals

Two completed charges can be compared only after identifying their translation spaces and calibrations. The projective metric of a Lorentz cone is standard [20]; its expression in null-response ratios separates relative mass scale from relative observer motion.

Proposition 3.2 (Response interval). *Let $P = m_P U$ and $Q = m_Q V$ be future timelike charges in the same translation space. If $a = \min_{S^2}(F_Q/F_P)$, $b = \max_{S^2}(F_Q/F_P)$, and $\chi = \text{arcosh}[-\eta(U, V)]$, then*

$$a = \frac{m_Q}{m_P} e^{-\chi}, \quad b = \frac{m_Q}{m_P} e^{\chi}, \quad \frac{m_Q}{m_P} = \sqrt{ab}, \quad \chi = \frac{1}{2} \log \frac{b}{a}. \quad (9)$$

The endpoints are invariant under simultaneous Lorentz transformations. Moreover, $Q - aP$ and $bP - Q$ are future null or zero. With the convention $d_{\text{Hilbert}} = \log(b/a)$, the projective distance is 2χ .

Proof. In the rest frame of P , write $V = (\cosh \chi, \sinh \chi v)$, with $|v| = 1$ when $\chi > 0$. The ratio is $(m_Q/m_P)(\cosh \chi - \sinh \chi v \cdot n)$, giving the extrema in (9). Its two limiting differences have nonnegative null responses with a zero minimum, and hence lie on the future cone boundary. Under (8), both responses acquire the same factor ω^{-1} , so their ratio is unchanged. $\square$

If an actual total charge is $P+Q$, its invariant mass obeys $m_{P+Q}^2 = m_P^2 + m_Q^2 + 2m_P m_Q \cosh \chi$. This is the ordinary relativistic composition law expressed through the boundary comparison interval. In gravity, identifying a sum of subsystem charges with the total ADM charge requires control of field and binding contributions; the interval itself supplies no such dynamical assumption.

3.3 Which angular responses close as one translation?

For a general smooth real function F on the reference round sphere, define

$$\mathcal{E}_F(U) = \frac{1}{4\pi} \int_{S^2} F(n) A_U(n)^{-3} d\Omega(n). \quad (10)$$

This is precisely the weight-three transform used in the null-to-timelike extension of asymptotic symmetry parameters [12]. For a translation response, Proposition 3.1 identifies the same transform with its Chern-weighted observer mean.

Theorem 3.3 (Translation-response closure). *Use $\Delta_H = \text{div } \nabla$ and let D be the round-sphere covariant derivative. Every smooth F satisfies $\Delta_H \mathcal{E}_F = 3\mathcal{E}_F$. The following stronger conditions are equivalent:*

$$F \in \mathcal{H}_0 \oplus \mathcal{H}_1, \quad \text{Hess}_H \mathcal{E}_F = \mathcal{E}_F g_H, \quad (D^2 F)^{\text{TF}} = 0. \quad (11)$$

Here $\mathcal{H}_\ell$ denotes the degree- ℓ spherical harmonics. Thus scalar hyperbolic extension alone does not identify a translation covector; the full Hessian condition does.

Proof. Since $\ell(n)$ is null, $\text{Hess}_H A_U = A_U g_H$ and $|\nabla_H A_U|^2 = A_U^2$. Hence $\Delta_H A_U^{-s} = s(s-2)A_U^{-s}$. Taking $s = 3$ and differentiating under the angular integral proves the scalar equation. Smoothness follows because A_U is bounded away from zero on compact observer sets.

For $U_r = (\cosh r, \sinh r \xi)$, direct integration gives $\langle A_{U_r}^{-3} \rangle = \cosh r$. Therefore $A_{U_r}^{-3}/(4\pi \cosh r)$ is a probability density. Its mass outside every fixed cap around ξ tends to zero, and uniform continuity gives

$$\lim_{r \rightarrow \infty} \frac{\mathcal{E}_F(U_r)}{\cosh r} = F(\xi). \quad (12)$$

The limit is uniform in ξ for continuous F , so the transform is injective.

The standard translation-mode integrals yield $\mathcal{E}_{b_0+b \cdot n}(U) = b_0 U^0 + b \cdot \mathbf{U}$, which satisfies the full Hessian equation. Conversely, match the value and tangent gradient of a solution of that equation at one observer by an ambient linear function $L(U)$. Along every geodesic from that observer, both functions solve $y'' = y$ with the same initial data. They agree throughout H^3 , and (12) gives $F = b_0 + b \cdot n$. Finally, the harmonic identity (14) below shows that the trace-free sphere Hessian has exactly the same kernel. $\square$

For a general response the Lorentz law is $F'(n') = F(n)/\omega(n)$, rather than ordinary scalar pullback. Together with (8), it makes (10) covariant. The four-dimensional closed sector in (11) is invariant under this weighted action. Positivity does not replace closure: $F = 1 + \varepsilon P_2(n_z)$, with $0 < \varepsilon < 1$ and $P_2(t) = (3t^2 - 1)/2$, has a positive transform satisfying the scalar equation, but at rest

$$(\text{Hess}_H \mathcal{E}_F - \mathcal{E}_F g_H)_{(1,0)} = \frac{4\varepsilon}{5} \text{diag}(-1, -1, 2). \quad (13)$$

3.4 A sharp estimate for imperfect closure

The exact criterion admits a quantitative version in a fixed calibrated round frame. Let $\Pi_{\leq 1}$ be the round L^2 projection onto $\mathcal{H}_0 \oplus \mathcal{H}_1$, put $r = F - \Pi_{\leq 1} F$, and define $\mathcal{D}(F) = \int_{S^2} |(D^2 F)^{\text{TF}}|^2 d\Omega$.

Proposition 3.4 (Closure defect). *With orthonormal harmonics satisfying $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}$, one has*

$$\mathcal{D}(F) = \frac{1}{2} \sum_{\ell \geq 2, m} (\ell-1)\ell(\ell+1)(\ell+2) |F_{\ell m}|^2, \quad (14)$$

$$\|r\|_{L^2}^2 \leq \frac{\mathcal{D}(F)}{12}, \quad \|r\|_{\infty} \leq \sqrt{\frac{\mathcal{D}(F)}{6\pi}}. \quad (15)$$

Both constants are sharp. If $\Pi_{\leq 1} F = E - p \cdot n$ and $\delta_F = \sqrt{\mathcal{D}(F)/(6\pi)}$, then $E - |p| \geq \min_{S^2} F - \delta_F$. In particular, $\min F > \delta_F$ guarantees a future timelike translation fit; conversely, $E - |p| > \delta_F$ guarantees $F > 0$.

Proof. The integrated Bochner identity on the unit sphere is $\int |D^2 F|^2 = \int (\Delta F)^2 - \int |DF|^2$. Subtracting half the trace square gives (14). Its smallest positive coefficient is 12, attained at $\ell = 2$, proving the sharp L^2 bound. Write $\lambda_\ell = (\ell-1)\ell(\ell+1)(\ell+2)/2$. Cauchy–Schwarz and the addition theorem give

$$|r(n)|^2 \leq \mathcal{D}(F) \sum_{\ell \geq 2} \frac{2\ell+1}{4\pi\lambda_\ell} = \frac{\mathcal{D}(F)}{6\pi}. \quad (16)$$

The sum follows from $(2\ell+1)/[(\ell-1)\ell(\ell+1)(\ell+2)] = \frac{1}{2}\{[(\ell-1)\ell]^{-1} - [(\ell+1)(\ell+2)]^{-1}\}$. Sharpness is approached by the smooth finite sums $r_L(n) = \sum_{\ell=2}^L (2\ell+1)P_\ell(n \cdot a)/(4\pi\lambda_\ell)$: the ratio $r_L(a)^2/\mathcal{D}(r_L)$ tends to $1/(6\pi)$. The cone statements follow by taking minima in $F = E - p \cdot n + r$. $\square$

The spectrum, Bochner identity, and reproducing-kernel argument are standard mathematical tools; the explicit estimate here quantifies their application to MS response closure without a claim of historical priority. For a nonclosed F , both the numerical defect and its translation projection depend on the reference observer. That projection is not a canonically defined ADM covector. Exact closure, by contrast, selects a Lorentz-invariant four-dimensional response space.

These results identify three descriptions of the same completed translation data: a positive Hermitian form tested by null directions, angular response moments calibrated to an observer, and a static potential on observer space. The sphere of null probes remains distinct from the spatial sphere on which an ADM surface charge is evaluated. Their connection is through the completed charge and its dual pairing, rather than an identification of the two boundaries. The correspondence constrains response geometry but does not determine the gravitational coupling or reconstruct the intervening gravitational field.

4 Classical positive-response identities

4.1 Conventions and asymptotic charges

Let (Σ, h, k) be a smooth asymptotically flat initial data set with future unit normal N . We use signature $(-, +, +, +)$ and the convention $k = -K$ relative to the extrinsic curvature of the companion boundary-charge paper. Write $\tau = \text{tr}_h k$, $\epsilon = T(N, N)$ and $j_i = -T(N, \partial_i)$. Thus j_i is a covector and $j^i = h^{ij} j_j$. The Einstein constraints, with $\kappa = 8\pi G > 0$ and $c = 1$, are

$$R_h + \tau^2 - |k|_h^2 = 2\kappa\epsilon, \quad D_j(k^{ij} - \tau h^{ij}) = \kappa j^i. \quad (17)$$

We assume the dominant energy condition (DEC), $\epsilon \geq |j|_h$. In an orthonormal spin frame this is equivalent to $\epsilon \mathbf{1} - j_a \sigma_a \geq 0$.

In asymptotically Cartesian coordinates the calibrated translation charges are

$$E = \frac{1}{2\kappa} \lim_{r \rightarrow \infty} \int_{S_r} (\partial_j h_{ij} - \partial_i h_{jj}) \nu^i dS, \quad p_i = \frac{1}{\kappa} \lim_{r \rightarrow \infty} \int_{S_r} (k_{ij} - \tau h_{ij}) \nu_h^j dS_h. \quad (18)$$

Here ν and dS are Euclidean, whereas ν_h and dS_h are determined by h . Their limits are the standard ADM charges with the stated convention for k . We retain $F(n) = E - p \cdot n$ as the response to a calibrated asymptotic null direction n .

4.2 A conformally flat maximal identity

Assume first that $\Sigma = \mathbb{R}^3$ has no inner boundary and

$$h_{ij} = \lambda^2 \delta_{ij} = \psi^4 \delta_{ij}, \quad \lambda = \psi^2 > 0, \quad \tau = 0. \quad (19)$$

Euclidean derivatives, contractions and norms are used in this subsection unless indicated otherwise. Put $A^{ij} = \lambda^5 k^{ij}$ and lower the indices of A with δ . The tensor A is Euclidean trace-free. The constraints become

$$\Delta \lambda = -\frac{1}{4} \lambda^{-3} |A|^2 - \frac{\kappa}{2} \epsilon \lambda^3 + \frac{1}{2\lambda} |\nabla \lambda|^2, \quad \partial_j A^{ij} = \kappa \lambda^5 j^i. \quad (20)$$

In the momentum equation j^i is still the physical contravariant current, not the Euclidean raising of j_i .

Suppose $\lambda = 1 + \sigma/r + O_2(r^{-2})$, $A = O_1(r^{-2})$, and $\epsilon, |j|_h$ are integrable with respect to dV_h . Then

$$E = -\frac{2}{\kappa} \lim_{r \rightarrow \infty} \int_{S_r} \partial_\nu \lambda dS = \frac{8\pi\sigma}{\kappa}, \quad p_i = \frac{1}{\kappa} \lim_{r \rightarrow \infty} \int_{S_r} A_{ij} \nu^j dS. \quad (21)$$

For a Euclidean unit vector n , define $\mathbf{S}(n, v)_{ij} = \frac{1}{2}(n_i v_j + n_j v_i) - \frac{1}{3}(n \cdot v) \delta_{ij}$ and $\mathcal{F}_n = -2\lambda^{-2} \nabla \lambda - \lambda^{-3} A n$.

Lemma 4.1 (Direct constraint identity). *Under (20),*

$$\begin{aligned} \operatorname{div} \mathcal{F}_n = & \frac{1}{2} \lambda^{-5} |A + 3\lambda \mathbf{S}(n, \nabla \lambda)|^2 + \frac{3}{4} \lambda^{-3} (|\nabla \lambda|^2 - (n \cdot \nabla \lambda)^2) \\ & + \kappa \lambda (\epsilon - \lambda^{-1} n^i j_i). \end{aligned} \quad (22)$$

The right-hand side is nonnegative under DEC.

Proof. The scalar equation gives

$$\operatorname{div}(-2\lambda^{-2} \nabla \lambda) = \frac{1}{2} \lambda^{-5} |A|^2 + 3\lambda^{-3} |\nabla \lambda|^2 + \kappa \lambda \epsilon.$$

Since $j^i = \lambda^{-2} \delta^{ij} j_j$, the momentum equation gives

$$\operatorname{div}(-\lambda^{-3} A n) = 3\lambda^{-4} A(n, \nabla \lambda) - \kappa n^i j_i.$$

Now $A : \mathbf{S}(n, v) = A(n, v)$ and $|\mathbf{S}(n, v)|^2 = \frac{1}{2} |v|^2 + \frac{1}{6} (n \cdot v)^2$; completing the square proves (22). Finally $|j|_h = \lambda^{-1} |j|_\delta$, so $\lambda^{-1} n^i j_i \leq |j|_h \leq \epsilon$. $\square$

Theorem 4.2 (Positive energy in the conformally flat maximal class). *Let smooth data on $\mathbb{R}^3$ satisfy (19), the stated asymptotics and integrability, the Einstein constraints and DEC. Then $E \geq |p|$. Equality implies $\lambda = 1$, $k = 0$, $\epsilon = j = 0$ and $E = |p| = 0$.*

Proof. Smooth positivity and the asymptotic limit of λ give uniform upper and lower bounds on λ . All the terms in (22) are therefore integrable. The divergence theorem and (21) give

$$\begin{aligned} E - p \cdot n = & \frac{1}{2\kappa} \int_{\mathbb{R}^3} \lambda^{-5} |A + 3\lambda \mathbf{S}(n, \nabla \lambda)|^2 d^3x \\ & + \frac{3}{4\kappa} \int_{\mathbb{R}^3} \lambda^{-3} |\nabla \lambda - (n \cdot \nabla \lambda) n|^2 d^3x + \int_{\mathbb{R}^3} \lambda (\epsilon - \lambda^{-1} n^i j_i) d^3x. \end{aligned} \quad (23)$$

This proves $F(n) \geq 0$ for every n , and hence $E \geq |p|$. If equality holds, choose $n = p/|p|$ when $p \neq 0$ and any n otherwise. The transverse gradient vanishes. For each point, moving to infinity along a line perpendicular to n then gives $\lambda = 1$. The first square forces $A = 0$, the scalar constraint gives $\epsilon = 0$, and DEC gives $j = 0$. $\square$

The formula also holds on punctured domains with their inner fluxes retained. To remove those terms one must check their vanishing at each end; it does not follow merely from the existence of a puncture. Standard Bowen–York and Brill–Lindquist constructions provide familiar data on which these conditions can be checked [21–23].

The matter weight has a direct response interpretation. The lapse $L = \lambda^{-2}$ and shift $Y^i = \lambda^{-3} n^i$ define the future-null field $\ell_n = LN + Y$, since $|Y|_h = L$. Moreover, $T(N, \ell_n) dV_h = (\lambda \epsilon - n^i j_i) d^3x$, which is precisely the matter term in (23). The conformal formula is an explicit positive decomposition in this restricted class; the general positive-mass theorem is established independently [6, 7].

4.3 The obstruction to the same conformal calculation

The preceding square completion does not extend unchanged to a nonmaximal slice. If $A = \lambda^5 (k - \frac{1}{3} \tau h)^{ij}$, the scalar equation acquires $\lambda^3 \tau^2 / 6$ and the momentum equation acquires $2\lambda^3 \partial^i \tau / 3$. Adding $2\tau n / 3$ to $\mathcal{F}_n$ cancels the derivative of τ in its divergence, but leaves the negative term $-\lambda \tau^2 / 3$. A nonflat reference metric in $h = \psi^4 \bar{h}$ likewise introduces reference-curvature and derivative contributions without a sign supplied by DEC.

These are obstructions to this particular current, rather than to positivity itself. Spacetime-harmonic functions give an established alternative after level-set curvature and topology have been controlled [24, 25]. We instead examine a frame current, for which the relation to the spinorial proof can be stated exactly.

4.4 An orthonormal-frame current

Let (e_1, e_2, e_3) be an oriented h -orthonormal frame. Define

$$\Gamma_{abc} = h(D_{e_a} e_b, e_c) = \epsilon_{bcd} M_{ad}, \quad d_a = \operatorname{div}_h e_a = \epsilon_{abc} M_{bc}. \quad (24)$$

Put $t = \operatorname{tr} M$, $H = \operatorname{TF} \operatorname{sym} M$ and $C = -2 \sum_a d_a e_a$. The scalar curvature identity is

$$R_h = \operatorname{div}_h C + \frac{1}{2}|d|^2 - |H|^2 + \frac{2}{3}t^2. \quad (25)$$

This identity uses no reference metric. Its current still depends on the chosen frame; coordinate covariance does not make it invariant under arbitrary local frame rotations.

Choose $a = e_3$, let $L > 0$, and set $u = \nabla_h \log L$. The conditions to be solved are

$$d_a = -e_a(\log L) + \tau \delta_{a3}, \quad t = 0. \quad (26)$$

For frame components define

$$\begin{aligned} S(a, u)_{ij} &= \frac{1}{2}(a_i u_j + a_j u_i) - \frac{1}{3}u(a)\delta_{ij}, \\ U_a(H)_{ij} &= \frac{1}{2}(\epsilon_{3jd} H_{id} + \epsilon_{3id} H_{jd}), \end{aligned} \quad (27)$$

and put $k_0 = k - \frac{1}{3}\tau h$. The current is

$$J = \frac{L}{2}C - L(k - \tau h)(a, \cdot)^\sharp. \quad (28)$$

Proposition 4.3 (Conditional frame-current identity). *If the Einstein constraints and (26) hold, then*

$$\begin{aligned} \operatorname{div}_h J &= L \left\{ \frac{1}{2} \left| k_0 - U_a(H) - \frac{3}{2}S(a, u) + \frac{\tau}{2}S(a, a) \right|^2 \right. \\ &\quad \left. + \frac{3}{4} \left[H(a, a)^2 + \left| H(a, \cdot)^\sharp - H(a, a)a - \frac{1}{2}a \times u \right|^2 \right] + \kappa(\epsilon - j(a)) \right\}. \end{aligned} \quad (29)$$

The right-hand side is nonnegative under DEC, with no restriction on τ .

Proof. With $t = 0$, write $M_{ij} = H_{ij} + \frac{1}{2}\epsilon_{ijc}d_c$. The symmetric derivative of a is

$$\operatorname{sym}(Da) = U_a(H) - \frac{1}{2}S(a, d) + \frac{1}{3}d(a)h.$$

Differentiate (28), insert (25) and the constraints, and use $d = -u + \tau a$. The result is

$$\begin{aligned} \frac{\operatorname{div}_h J}{L} &= \frac{1}{2}|k_0|^2 - k_0 : \left(U_a(H) + \frac{3}{2}S(a, u) - \frac{\tau}{2}S(a, a) \right) \\ &\quad + \frac{1}{2}|H|^2 + \frac{3}{4}|u|^2 + \frac{\tau^2}{12} - \frac{\tau}{2}u(a) + \kappa(\epsilon - j(a)). \end{aligned}$$

The contractions $U_a(H) : S(a, a) = 0$, $S(a, u) : S(a, a) = 2u(a)/3$ and $|S(a, a)|^2 = 2/3$ absorb the trace terms into the first square of (29). The remaining quadratic form is

$$\frac{1}{2}|H|^2 + \frac{3}{4}|u|^2 - \frac{1}{2} \left| U_a(H) + \frac{3}{2}S(a, u) \right|^2 = \frac{3}{4} \left[H(a, a)^2 + \left| H(a)_\perp - \frac{1}{2}a \times u \right|^2 \right].$$

This can be checked in the frame $a = e_3$ by writing a general symmetric trace-free matrix H , and proves the identity. $\square$

In this gauge $J = \nabla_h L - Lk(a, \cdot)^\sharp$, and $L(N + a)$ is a future-null probe. Its matter response is $L(\epsilon - j(a))$.

For the boundary identification suppose $h - \delta = O_2(r^{-q})$, $k = O_1(r^{-q-1})$, $q > 1/2$, the frame is asymptotically Cartesian with first derivatives of the corresponding order, $L = 1 + O_1(r^{-q})$ and $a = n + O_1(r^{-q})$. Assume the limits (18) exist. Then

$$\frac{1}{\kappa} \lim_{r \rightarrow \infty} \int_{S_r} h(J, \nu_h) dS_h = E - p \cdot n. \quad (30)$$

To see the energy term explicitly, write $h_{ij} = \delta_{ij} + s_{ij}$ and $e_a = (\delta_a^i + b_a^i) \partial_i$ at the selected end. Orthonormality gives $b_{ij} = -s_{ij}/2 + \omega_{ij} + O(r^{-2q})$, with $\omega_{ij} = -\omega_{ji}$. Thus $C^i/2 = (\partial_j s_{ij} - \partial_i s_{jj})/2 - \partial_j \omega_{ij} + O(r^{-2q-1})$. The antisymmetric term has zero flux through a closed sphere; the error has flux $O(r^{1-2q})$. The lapse, direction and metric corrections to the momentum term have the same vanishing order. This proves (30). Inner boundaries would supply additional fluxes and require separate boundary conditions.

4.5 Perturbative solvability of the frame gauge

For precision, use weighted Hölder spaces on $\mathbb{R}^3$ in which $v \in C_{-q}^{r, \alpha}$ means $\partial^\ell v = O((1 + |x|)^{-q-\ell})$, with the corresponding weighted Hölder seminorm. Fix an integer $m \geq 2$, $0 < \alpha < 1$ and $1/2 < q < 1$. The relevant domain and target for a first-order gauge map are $C_{-q}^{m+1, \alpha}(\mathbb{R}^3; \mathbb{R}^4)$ and $C_{-q-1}^{m, \alpha}(\mathbb{R}^3; \mathbb{R}^4)$.

Let $e_a^0 = (h^{-1/2})_a^i \partial_i$ be the symmetric square-root frame, set $e_a = (\exp \Omega(\theta))_a^b e_b^0$ with $\Omega(\theta)_{ab} = \epsilon_{abc} \theta_c$, and write $L = e^f$. In a fixed asymptotic orientation the gauge map is

$$\mathcal{G}_{h, \tau}(\theta, f) = (d_a + e_a f - \tau \delta_{a3}, \text{tr } M).$$

It is smooth between the stated spaces near Euclidean data. Its derivative at $(h, \tau, \theta, f) = (\delta, 0, 0, 0)$ is

$$\mathcal{L}_0(\theta, f) = (\text{curl } \theta + \nabla f, \text{div } \theta). \quad (31)$$

Lemma 4.4 (Weighted div-curl inverse). *The operator (31) is a bounded isomorphism from $C_{-q}^{m+1, \alpha}(\mathbb{R}^3; \mathbb{R}^4)$ to $C_{-q-1}^{m, \alpha}(\mathbb{R}^3; \mathbb{R}^4)$ for $0 < q < 1$.*

Proof. For prescribed (F, g) solve $\Delta f = \text{div } F$ and $\Delta \theta = \nabla g - \text{curl } F$. The Euclidean Laplacian is a bounded isomorphism $C_{-q}^{m+1, \alpha} \rightarrow C_{-q-2}^{m-1, \alpha}$ for $0 < q < 1$. The inverse and its weighted Schauder estimate follow from the Newton potential; this is the usual nonexceptional weight range for asymptotic elliptic estimates [26]. Put $v = \text{div } \theta - g$. It is harmonic, smooth on $\mathbb{R}^3$ and decays, hence vanishes. The field $w = \text{curl } \theta + \nabla f - F$ then has both zero divergence and zero curl. Its components are decaying harmonic functions, so $w = 0$ as well. The Laplacian estimates give the bounded inverse. The same argument with $(F, g) = 0$ proves injectivity. $\square$

Proposition 4.5 (Small-data frame gauge). *There is a neighborhood of $(\delta, 0)$ in $C_{-q}^{m+1, \alpha} \times C_{-q-1}^{m, \alpha}$ such that, if $(h - \delta, \tau)$ lies in that neighborhood, (26) has a global solution with $\theta, f \in C_{-q}^{m+1, \alpha}$ and $L = e^f > 0$. The statement holds for every fixed asymptotic direction $a \rightarrow n$, with a smallness threshold independent of n .*

Proof. Apply the Banach implicit-function theorem to $\mathcal{G}$, using Lemma 4.4. Positivity of L is automatic. A constant rotation of the asymptotic frame realizes any n ; Euclidean rotational invariance and compactness of $SO(3)$ give a uniform threshold. This does not require choosing a globally smooth transverse basis over the direction sphere. $\square$

Theorem 4.6 (Small-data energy–momentum inequality). *Let smooth complete Einstein initial data on $\mathbb{R}^3$, without an inner boundary, satisfy DEC and have well-defined ADM charges. Suppose $h - \delta \in C_{-q}^{m+1, \alpha}$ and $\tau \in C_{-q-1}^{m, \alpha}$ have the small norms of Proposition 4.5, while $k \in C_{-q-1}^{m, \alpha}$ and $\epsilon, |j|_h \in L^1(dV_h)$. Then $E \geq |p|$. Conformal flatness and maximality are not required.*

Proof. For each fixed n , obtain the global gauge from Proposition 4.5. Its first derivatives, the connection coefficients and k decay as r^{-q-1} , so the quadratic terms in (29) are integrable because $2q > 1$. The matter term is integrable as well. Integrating the positive current and using (30) gives $F(n) \geq 0$. Cone duality gives the conclusion. $\square$

This is a perturbative construction of the particular frame gauge. It supplies neither its continuation to arbitrary field strength nor a replacement for the global positive-mass theorem and its rigidity analysis. The trace-free part of k enters the identity and decay assumptions, but does not enter the gauge map; it need not have a small norm for this conditional argument.

5 The Sen–Witten lift and what positivity selects

5.1 The frame equation and its spinorial meaning

Let σ_a be Hermitian Pauli matrices in the chosen spin frame. The intrinsic spin connection is

$$\nabla_a^h \chi = e_a \chi - \frac{i}{2} M_{ad} \sigma_d \chi. \quad (32)$$

On a region where the spinor is nonzero, an adapted spin frame can write $\chi = \sqrt{L} \chi_0$, with χ_0 constant, $|\chi_0| = 1$ and $\sigma_3 \chi_0 = \chi_0$. Then

$$\sigma^a \nabla_a^h \chi = \frac{1}{2} [(u + d)_a \sigma_a - it] \chi. \quad (33)$$

With the chosen sign of k , the Sen connection and contracted equation are

$$\nabla_a^{\text{Sen}} \chi = \nabla_a^h \chi - \frac{1}{2} k_{ab} \sigma_b \chi, \quad D_{\text{Sen}} \chi := \sigma^a \nabla_a^{\text{Sen}} \chi = \left(\sigma^a \nabla_a^h - \frac{\tau}{2} \right) \chi. \quad (34)$$

Symmetry of k removes the antisymmetric Pauli product. Since $(a \cdot \sigma) \chi = \chi$, the equation $D_{\text{Sen}} \chi = 0$ is equivalent in the adapted frame to the four real conditions (26).

The relation between Witten spinors and special orthonormal frames is established [7, 9, 10, 23, 27]. The preceding square completion lies within that mechanism. A boundary null direction supplies a projective spinor line; solving the Sen–Witten equation also requires its spinorial lift and the global elliptic problem. An adapted-frame description cannot be used across a spinor zero without additional work.

5.2 A current that remains regular at spinor zeros

The spinorial identity avoids that restriction. For a smooth spinor satisfying $D_{\text{Sen}} \chi = 0$, put $L = \chi^\dagger \chi$, $v_a = \chi^\dagger \sigma_a \chi$ and

$$J_\chi = \nabla_h L - k(v, \cdot)^\sharp. \quad (35)$$

This is a global polynomial expression in χ and its derivatives; it contains no division by L .

Proposition 5.1 (Spinorial positive-current identity). *Under the Einstein constraints,*

$$\text{div}_h J_\chi = 2 \sum_a |\nabla_a^{\text{Sen}} \chi|^2 + \kappa \chi^\dagger (\epsilon \mathbf{1} - j_a \sigma_a) \chi. \quad (36)$$

Under DEC the right-hand side is nonnegative, including at zeros of χ .

Proof. In the Hermitian-Pauli convention $D_0 = \sigma^a \nabla_a^h$ satisfies $D_0^2 = \Delta_{\text{spin}} - R_h/4$, where Δ_{spin} has Euclidean sign $\sum_i \partial_i^2$. Since $D_0 \chi = \tau \chi/2$,

$$\Delta_{\text{spin}} \chi = \frac{1}{4}(R_h + \tau^2) \chi + \frac{1}{2} \sigma(\text{d}\tau) \chi, \quad \Delta_h L = 2|\nabla^h \chi|^2 + \frac{1}{2}(R_h + \tau^2)L + \text{d}\tau \cdot v.$$

The momentum constraint gives

$$\text{div}_h(k(v, \cdot)^\sharp) = (\kappa j + \text{d}\tau) \cdot v + 2 \text{Re} \sum_{a,b} k_{ab} \chi^\dagger \sigma_b \nabla_a^h \chi.$$

Expansion of the Sen norm yields

$$2|\nabla^{\text{Sen}} \chi|^2 = 2|\nabla^h \chi|^2 - 2 \text{Re} \sum_{a,b} k_{ab} \chi^\dagger \sigma_b \nabla_a^h \chi + \frac{1}{2}|k|^2 L.$$

Subtract the divergence above from $\Delta_h L$, use the scalar constraint, and obtain (36). These calculations are pointwise polynomial identities and remain valid when $\chi = 0$. $\square$

This is the standard Witten–Sen positivity mechanism in the present current and sign conventions, rather than a new positive-mass theorem. On a complete spin initial-data slice its use with a prescribed asymptotic spinor requires the established existence, decay and boundary analysis [7, 8]. On a slice with an inner boundary, the corresponding boundary term must also be controlled. Its polynomial form will allow several asymptotic spinor inputs to be polarized without assuming that any one adapted frame exists globally.

5.3 A restricted coefficient-selection test

The spacetime spin restriction fixes the factor $1/2$ in (34). Positivity gives a useful, but more limited, test of the same factor. Keep the current (28) fixed and replace its gauge by

$$d = -u + \gamma \tau a, \quad t = 0. \quad (37)$$

In an adapted spin frame this is the equation $D_b \chi = 0$, with $D_b = \sigma^a \nabla_a^h - b\tau$ and $\gamma = 2b$.

Proposition 5.2 (Restricted algebraic coefficient test). *Under the constraints and (37),*

$$\begin{aligned} \frac{\text{div}_h J}{L} &= \frac{1}{2} \left| k_0 - \text{U}_a(H) - \frac{3}{2} \text{S}(a, u) + \frac{\gamma \tau}{2} \text{S}(a, a) \right|^2 \\ &\quad + \frac{3}{4} \left[H(a, a)^2 + \left| H(a)_\perp - \frac{1}{2} a \times u \right|^2 \right] + \kappa(\epsilon - j(a)) - \frac{(\gamma - 1)^2}{3} \tau^2. \end{aligned} \quad (38)$$

Within this current and gauge family, a universal nonnegative algebraic expression for arbitrary local coefficient values subject to DEC requires $\gamma = 1$.

Proof. Before completing the square the terms independent of k_0 are

$$\frac{1}{2}|H|^2 + \frac{3}{4}|u|^2 - \frac{\gamma}{2}\tau u(a) + \left(-\frac{1}{3} - \frac{\gamma^2}{4} + \frac{2\gamma}{3} \right) \tau^2 + \kappa(\epsilon - j(a)).$$

The first square in (38) contributes $\gamma^2 \tau^2/12 - \gamma \tau u(a)/2$; the remainder is $-(\gamma - 1)^2 \tau^2/3$. For $\gamma \neq 1$, set $H = u = 0$, choose $\tau \neq 0$, and set $k_0 = -\gamma \tau \text{S}(a, a)/2$ and $\epsilon = j = 0$ in this algebraic expression. Both squares vanish and the remainder is strictly negative. $\square$

The last choice tests the reduced pointwise quadratic form. It is not a construction of global constraint data or a counterexample to the positive-mass theorem. Nor does the test classify all covariant gauges or derive the Sen equation from MS response data. Its conclusion concerns this fixed current and one-parameter ansatz; more general gauge choices for Witten energy expressions are established [11].

5.4 What phase-blind comparison cannot select

For the adapted spinor above, set $r = d + u - \tau a$. Equation (33) gives

$$D_{\text{Sen}}\chi = \frac{\sqrt{L}}{2}(r_a\sigma_a - it)\chi_0, \quad |D_{\text{Sen}}\chi|^2 = \frac{L}{4}(|r|^2 + t^2). \quad (39)$$

By contrast the phase-invariant null bilinears L and $v = La$ obey only the single real projection

$$2\operatorname{Re}(\chi^\dagger D_{\text{Sen}}\chi) = \operatorname{div}_h v - \tau L. \quad (40)$$

Thus the null-current balance $\operatorname{div}_h v = \tau L$ is one component of the spinor equation, not the whole equation.

Proposition 5.3 (Phase-blind comparison obstruction). *Fix h , k and the spin connection. A selection rule that depends on χ only through pointwise Hermitian bilinears $\chi^\dagger A\chi$ and derivatives of those phase-invariant bilinears cannot characterize precisely the Sen–Witten solutions with a prescribed constant asymptotic value.*

Proof. On Euclidean data with $k = 0$, let χ_0 be a constant unit spinor and f a nonconstant smooth real function of compact support. The spinors χ_0 and $\chi_f = e^{if}\chi_0$ have the same pointwise Hermitian bilinears, all derivatives of those bilinears, and asymptotic value. Nevertheless

$$D_{\text{Sen}}\chi_f = ie^{if}\sigma^a(\partial_a f)\chi_0, \quad |D_{\text{Sen}}\chi_f|^2 = |\nabla f|^2,$$

whereas $D_{\text{Sen}}\chi_0 = 0$. The stated rule cannot distinguish them. $\square$

The phase change in this argument keeps the connection fixed. It is not a passive change of spin frame accompanied by a transformation of the connection. Derivative bilinears such as $\chi^\dagger \nabla \chi$ can retain phase information and are not covered by the proposition. A phase-sensitive response algebra is therefore not excluded. Neither is a momentum-space helicity connection identified with the spatial connection selecting a Sen–Witten field.

For the same reason, the functional $\int_\Sigma |D_{\text{Sen}}\chi|^2 dV_h$ is not an independent selection theorem merely because it is nonnegative. Compactly supported variation gives $D_{\text{Sen}}^\dagger D_{\text{Sen}}\chi = 0$; attainment of a zero minimum with the required asymptotic value still needs the elliptic theory. That analytic input belongs to the established Witten/Parker–Taubes construction [7, 8], rather than following from projective null data alone.

6 The matrix Sen–Witten response lift

The scalar current can be polarized in the asymptotic spinor input. This retains all null probes in one linear construction and avoids the need for a nonvanishing adapted spinor in the interior.

Take smooth complete spin initial data with a chosen asymptotically flat end, no inner boundary, the constraints (17), DEC, and decay sufficient for the ADM flux and the following integrals. We assume the standard Sen–Witten boundary problem is uniquely solvable for every constant asymptotic spinor, with finite Sen-gradient norm and vanishing error terms in the limiting flux. These are the analytic hypotheses of the established spinorial positive-energy argument [7, 8]. For several ends, choose zero asymptotic spinor data at the other ends or retain their boundary terms explicitly.

Choose an orthonormal asymptotic spin basis e_1, e_2 . Let χ_A solve $D_{\text{Sen}}\chi_A = 0$ and approach e_A at the chosen end. The two columns define $\Psi(x) : \mathbb{C}^2 \rightarrow \mathbb{S}_x$, where $\mathbb{S}$ is the hypersurface spinor bundle. Linearity gives $\chi_\zeta = \Psi\zeta$. The matter matrix in an orthonormal frame is $M = \epsilon \mathbf{1} - j_a \sigma_a \geq 0$.

Proposition 6.1 (Polarized positive-energy identity). *Under these hypotheses,*

$$\mathbf{P} = \frac{2}{\kappa} \int_{\Sigma} \sum_a (\nabla_a^{\text{Sen}} \Psi)^\dagger \nabla_a^{\text{Sen}} \Psi \, dV_h + \int_{\Sigma} \Psi^\dagger M \Psi \, dV_h. \quad (41)$$

Proof. The scalar polynomial current of Section 5 gives the matrix current

$$\mathcal{J}_b = \nabla_b(\Psi^\dagger \Psi) - k_{ba} \Psi^\dagger \sigma_a \Psi.$$

Applying its scalar identity to every $\Psi\zeta$ and polarizing yields

$$\text{div}_h \mathcal{J} = 2 \sum_a (\nabla_a^{\text{Sen}} \Psi)^\dagger \nabla_a^{\text{Sen}} \Psi + \kappa \Psi^\dagger M \Psi.$$

The asymptotic flux is $\kappa \mathbf{P}$: its quadratic form at a unit ζ is $\kappa(E - p \cdot n)$, $n = \zeta^\dagger \sigma \zeta$. Integration proves (41). The current is polynomial in Ψ and its derivatives, so zeros of a column or of $\det \Psi$ cause no singularity. $\square$

Define a response Hilbert space

$$\mathcal{R} = L^2(T^* \Sigma \otimes \mathbb{S}) \oplus L^2(\mathbb{S}), \quad T\zeta = \left(\sqrt{2/\kappa} \nabla^{\text{Sen}}(\Psi\zeta), M^{1/2} \Psi\zeta \right).$$

Then (41) is $\mathbf{P} = T^\dagger T$. Its sesquilinear response kernel is $K(\zeta, \eta) = \langle T\zeta, T\eta \rangle = \zeta^\dagger \mathbf{P} \eta$. For a unit input, $F_P(n) = \|T\zeta\|^2$; for every positive normalized MS probe ρ , $\text{Tr}(\mathbf{P}\rho) = \text{Tr}(T\rho T^\dagger)$. The latter congruence is mathematically completely positive. It is an auxiliary classical spinorial construction, not a physical quantum-state map.

Corollary 6.2 (Mass as response-channel area). *Put $t_A = Te_A$ and normalize the exterior product so that orthonormal vectors have unit wedge norm. Then*

$$m^2 = \det \mathbf{P} = \|t_1\|^2 \|t_2\|^2 - |\langle t_1, t_2 \rangle|^2 = \|t_1 \wedge t_2\|^2. \quad (42)$$

Together with the observer moments of Section 3, this gives

$$m^2 = \mathcal{E}_P(U)^2 - 3 \text{Var}_{\mu_U}(F_P/A_U) = \|Te_1 \wedge Te_2\|^2. \quad (43)$$

Proof. Equation (42) is the determinant formula for the Gram matrix of two vectors. The other equality follows from the uniform second moment of the observer's null sphere. $\square$

For local orthonormal response components $t_{\mu A}(x)$, continuous Cauchy–Binet makes the area explicit:

$$m^2 = \frac{1}{2} \int_{\Sigma} \int_{\Sigma} \sum_{\mu, \nu} |t_{\mu 1}(x) t_{\nu 2}(y) - t_{\mu 2}(x) t_{\nu 1}(y)|^2 \, dV_x dV_y. \quad (44)$$

Expanding the square gives twice the Gram determinant. The sum is invariant under independent unitary choices of local response basis. The associated positive integrand of (41) is not a uniquely defined local gravitational stress tensor. Only its integrated boundary charge has the ADM identification.

The rank of T equals the rank of $\mathbf{P}$. Algebraically, rank two corresponds to a timelike charge and rank one to a nonzero null charge. Actual equality cases for complete initial data are subject to the usual positive-mass rigidity hypotheses; the rank-one kinematical boundary should not be taken as a generic regular ADM state. Also, $\mathbf{P}$ does not determine Ψ or T : the elliptic equation and the bulk geometry remain essential. Under phase changes of the two probe lifts, the off-diagonal kernel transforms as a line-valued object. Its diagonal descends to $\mathbb{CP}^1$, but its local representatives do not supply an independent phase-selection principle.

7 Boundary line metrics, curvature and spin representations

7.1 The response metric on the basic spinor line

Let $\mathcal{H} = \mathcal{O}(-1) \rightarrow \mathbb{CP}^1$ be the tautological line, oriented so that $c_1(\mathcal{H}) = -1$. A positive timelike response form supplies the fiber metric

$$h_P(v, v) = v^\dagger P v, \quad v \in \mathcal{H}_{[\zeta]}. \quad (45)$$

For a unit lift ζ of n , this is $F_P(n)$. Choose the real Chern-curvature convention $f = -i\partial\bar{\partial}\log H$ for the squared norm H of a holomorphic frame.

Proposition 7.1 (Response curvature). *For $P > 0$, the basic curvature and its helicity powers are*

$$f_{1/2,P} = -\frac{1}{2} \frac{m^2}{F_P(n)^2} d\Omega, \quad f_{h,P} = -h \frac{m^2}{F_P(n)^2} d\Omega, \quad c_1(L_h) = -2h, \quad (46)$$

where $h \in \frac{1}{2}\mathbb{Z}$ and $L_h \simeq \mathcal{H}^{\otimes 2h}$, with dual powers for negative $2h$.

Proof. In the affine chart $s(z) = (1, z)^T$, put $H_P = s^\dagger P s$. Differentiation gives $\partial_z \partial_{\bar{z}} \log H_P = \det P / H_P^2$. Since $F_P = H_P / (1 + |z|^2)$ and $d\Omega = 2i dz \wedge d\bar{z} / (1 + |z|^2)^2$, the basic formula follows. Tensor powers multiply the curvature; the total area below is 4π , giving the Chern number. $\square$

The corresponding visual metric and area form are

$$g_P = \frac{m^2}{F_P^2} g_{S^2}, \quad dA_P = \frac{m^2}{F_P^2} d\Omega. \quad (47)$$

The projective transformation $[\zeta] \mapsto [P^{1/2}\zeta]$ pulls the round metric and area back to (47). In particular, g_P has Gaussian curvature one and area 4π . The normalized measure $dA_P / (4\pi)$ is exactly μ_V for $V = P/m$ in Section 3. The Chern geometry therefore supplies the measure entering the calibrated observer mean.

The metric is unchanged by $P \mapsto \alpha P$, $\alpha > 0$, although the fiber norm scales by α . Curvature retains the projective timelike direction and loses the radial charge scale. This distinction persists for all tensor powers and is relevant to any attempt to determine the coupling normalization.

7.2 Curvature concentration and response-channel geometry

Set $D_P = p/E$, $S_P = m/E$ and $\beta = |D_P|$. The normalized Chern measure has density $S_P^2 / [4\pi(1 - D_P \cdot n)^2]$. For a cap of opening angle θ about D_P , its weight is

$$\mu_P(\text{cap}_\theta) = \frac{(1 + \beta)(1 - \cos \theta)}{2(1 - \beta \cos \theta)}, \quad \theta_{1/2} = \arccos \beta = \arcsin S_P. \quad (48)$$

This follows by integrating the density in $\cos \theta$. At rest the measure is uniform. If $\beta \rightarrow 1$ and $D_P / |D_P| \rightarrow n_*$, the measure converges weakly to a point mass at n_* ; the smooth positive line metric has no nondegenerate limit there. Vanishing determinant off the singular direction does not erase the Chern class of the timelike family.

For the Sen map of Section 6, h_P is the pullback of the response-space Hilbert norm under $v \mapsto T v$. If $P > 0$, T is injective and $[\zeta] \mapsto [T\zeta]$ is a holomorphic projective embedding. Its curvature is the pullback of the tautological curvature, equivalently minus the Fubini–Study form in the present normalization. Explicitly,

$$F_P(n) = \|T\zeta\|^2, \quad f_{1/2,P} = -\frac{\|Te_1 \wedge Te_2\|^2}{2\|T\zeta\|^4} d\Omega \quad (\|\zeta\| = 1). \quad (49)$$

The positive response norm, invariant determinant and curvature weight thus follow from the same Gram map. This is induced projective geometry; it does not equate the auxiliary Sen response space with a physical one-particle Hilbert space.

7.3 Symmetric powers and global section norms

The basic form also determines the higher-spin response metrics. Let $k = 2j \in \mathbb{Z}_{\geq 0}$ and $V_k = \text{Sym}^k \mathbb{C}^2$. Write $P^{\odot k}$ for the restriction of $P^{\otimes k}$ to V_k . For unit lifts put $\Pi_\zeta^{(k)} = |\zeta^{\otimes k}\rangle\langle\zeta^{\otimes k}|$.

Proposition 7.2 (Coherent response resolution). *The kernel $K_{P,k}(\zeta, \eta) = (\zeta^\dagger P \eta)^k$ is positive definite, with diagonal F_P^k , and*

$$\frac{k+1}{4\pi} \int_{S^2} \frac{\Pi_\zeta^{(k)}}{F_P(n)^k} dA_P = (P^{-1})^{\odot k}. \quad (50)$$

Proof. The kernel is the Gram pairing of $(P^{1/2}\zeta)^{\otimes k}$. Set $w = P^{1/2}\zeta/\sqrt{F_P}$, so dA_P is the pullback of $d\Omega_w$. The integrand equals $(P^{-1/2})^{\odot k} \Pi_w^{(k)} (P^{-1/2})^{\odot k}$. The integral of $\Pi_w^{(k)}$ commutes with the irreducible $SU(2)$ action. Its trace is 4π and its dimension is $k+1$, giving $4\pi \mathbf{1}/(k+1)$ by Schur's lemma. This is the standard coherent-state resolution [15, 28] in the response metric. $\square$

The inverse in (50) is essential. Holomorphic sections of the dual line $\mathcal{O}(k)$ are homogeneous polynomials $q_\alpha(z) = \alpha(z^{\otimes k})$, with $\alpha \in V_k^*$. Their normalized inner product is

$$\langle q_\alpha, q_\beta \rangle_P = \frac{k+1}{4\pi} \int_{\mathbb{CP}^1} \frac{\overline{q_\alpha(z)} q_\beta(z)}{(z^\dagger P z)^k} dA_P. \quad (51)$$

It is the dual form induced by $P^{\odot k}$. Borel–Weil identifies $H^0(\mathbb{CP}^1, \mathcal{O}(2j))$ with the spin- j representation, up to the conventional dual action, and gives dimension $2j+1$ [14]. The same response form therefore specifies the local line norm, Chern weight, coherent kernel and global section norm. These are representation metrics, with units of energy to the power k or its inverse; they are not new translation Hamiltonians.

Positive order is preserved by every symmetric power. If $P_1 \geq P_2 > 0$, then $P_1^{\odot k} \geq P_2^{\odot k}$ and $(P_1^{-1})^{\odot k} \leq (P_2^{-1})^{\odot k}$. Indeed, the tensor difference is a telescoping sum of positive tensor factors, and inverse order gives the second assertion. This will lift the Bondi response order to the coherent kernels without imposing pointwise order on their curvature densities.

7.4 Relation to helicity and Wigner transport

The helicity relation $c_1(L_h) = -2h$ is established independently in massless bundle representation theory [13, 29]. In the rest response form $P = m\mathbf{1}$, (46) reduces to its standard Berry curvature $-h d\Omega$. For general P , the response curvature is its pullback by the observer Möbius transformation. It need not equal the helicity-projector curvature in the unchanged reference angular coordinates.

The two metrics used in the companion spin construction should also be distinguished. A constant- S direction sphere inside H^3 has metric $(1 - S^2)S^{-2}g_{S^2}$ and Gaussian curvature $S^2/(1 - S^2)$. The visual metric (47) has Gaussian curvature one. Curvature dilution on the first and angular concentration on the second concern different metrics. Likewise, the canonical Wigner connection transports a spin carrier over the observer hyperboloid [30], whereas the Sen connection acts on a spinor field over the spatial initial-data manifold.

The bridge is the common projective spinor line and its induced representation geometry. In particular, the probe matrix has sign $\rho(D) = (1 + D \cdot \sigma)/2$, whereas a charge with $D_P = p/E$ has $P/(2E) = \rho(-D_P)$. At the aligned null limit the charge annihilates the probe line in direction D_P . This dual pairing must be retained; a physical helicity state is not thereby identified with a Sen–Witten test spinor.

8 Radiation and staticity balance

8.1 Positive null channels and invariant mass

Fix an asymptotic inertial calibration. Let μ be a finite nonnegative energy measure on the sphere of actual radiation directions ν , and define

$$R = \int_{S^2} (\mathbf{1} - \nu \cdot \sigma) d\mu(\nu) = e\mathbf{1} - r \cdot \sigma, \quad e = \mu(S^2), \quad r = \int \nu d\mu. \quad (52)$$

Each channel is null and positive. Consequently $|r| \leq e$, and the total radiation charge is future causal.

Proposition 8.1 (Angular response area). *For $e > 0$,*

$$\begin{aligned} m_R^2 &= e^2 - |r|^2 = \iint (1 - \nu \cdot \nu') d\mu(\nu) d\mu(\nu') \\ &= 2 \iint |\varepsilon(z_\nu, z_{\nu'})|^2 d\mu(\nu) d\mu(\nu'), \quad S_R^2 = \frac{m_R^2}{e^2} = \text{Var}_{\mu/e}(\nu). \end{aligned} \quad (53)$$

Here unit spinors represent $z_\nu z_\nu^\dagger = \Pi_{-\nu}$, and $\varepsilon(z, z') = z_1 z'_2 - z_2 z'_1$. The total charge is null exactly when μ is concentrated on one direction, up to null sets.

Proof. Expand $e^2 - |r|^2$ under the double integral. The Pauli projector identity gives $|\varepsilon(z_\nu, z_{\nu'})|^2 = (1 - \nu \cdot \nu')/2$. Finally, the unit norm of every ν gives $\text{Var}(\nu) = 1 - |r/e|^2$. Zero variance is equivalent to a constant direction almost everywhere. $\square$

The pairing modulus is globally defined although smooth spinor phases cannot be chosen over the entire sphere. These spinors encode null directions, not photon or graviton polarization states. Equation (53) is standard null-vector addition written in the same exterior-area language as (42). It concerns the total invariant mass of radiation: two opposite equal beams have nonzero total mass although every constituent is massless.

The two angular variances measure different things. The probe variance in (43) measures how unevenly one total charge responds to all test directions. The variance in (53) measures the spread of populated emission channels. For the same radiation charge in an observer U , they satisfy

$$\frac{3 \text{Var}_{\mu_U}(F_R/A_U)}{e_U^2} + \text{Var}_{\text{rad},U}(\nu_U) = 1. \quad (54)$$

To obtain the second probability measure in the U frame, aberrate directions, weight $d\mu$ by $A_U(\nu)$, and normalize the pushforward by $e_U = \mathcal{E}_R(U)$. A mere angular coordinate change would give the wrong identity.

8.2 Bondi translation balance

Let future null infinity be regular enough for the standard Bondi translation charges and flux law. In one fixed Bondi frame and cut calibration, assume a nonnegative outward flux density $\mathcal{L}(u, \nu)$. Then

$$\dot{E}_B = - \int \mathcal{L} d\Omega, \quad \dot{p}_B = - \int \nu \mathcal{L} d\Omega. \quad (55)$$

In vacuum we use the shear C_{AB} and news $N_{AB} = \partial_u C_{AB}$ in the convention of [31]. The flux density is $\mathcal{L} = (32\pi G)^{-1} N_{AB} N^{AB}$. An outward matter contribution can be added when its asymptotic flux law and positivity are supplied. Equation (55) is a dynamical input from general relativity, not a consequence of the MS cone [16–18].

For two cuts with $u_b < u_a$, let b denote the earlier charge and a the later charge. The balance gives $P_b = P_a + R$ with $d\mu = \int_{u_b}^{u_a} \mathcal{L} du d\Omega \geq 0$. Thus $P_b \geq P_a$. Every classical probe decreases:

$$-\dot{F}_B(n) = \int \mathcal{L}(u, \nu)(1 - n \cdot \nu) d\Omega \geq 0. \quad (56)$$

The kernel is $1 - n \cdot \nu = 2 \text{Tr}(\Pi_n \Pi_{-\nu})$. It pairs a test direction n with an emission direction ν ; the two spheres retain their distinct roles.

If the remaining Bondi charge is future causal, determinant polarization gives

$$m_b^2 = m_a^2 + m_R^2 + 2(E_a e - p_a \cdot r), \quad m_b \geq m_a + m_R. \quad (57)$$

The inequality is the reverse Lorentz triangle inequality. Bondi-charge positivity and any ADM–Bondi matching require their own regularity and energy assumptions. Cone order alone cannot supply those analytic results.

8.3 The MS staticity budget

For a future charge with $E > 0$, write $D = p/E$, $S = m/E$ and $q = m^2/E = E(1 - |D|^2)$. This is the same staticity quantity as in the massive-shell section when $c = 1$.

Proposition 8.2 (Finite and differential staticity balance). *For the Bondi balance above, with $E_a = a > 0$ and $e > 0$,*

$$q_b = q_a + q_R + \frac{ae}{a+e} |D_a - D_R|^2, \quad q_R = e \text{Var}_{\mu/e}(\nu). \quad (58)$$

At differentiable cuts with $E_B > 0$,

$$\dot{q}_B = - \int |\nu - D_B|^2 \mathcal{L} d\Omega \leq 0. \quad (59)$$

Proof. Use $D_b = (aD_a + eD_R)/(a+e)$ and expand the weighted variance in $q_b = (a+e)(1 - |D_b|^2)$. For the differential formula, put $\Phi = \int \mathcal{L} d\Omega$ and $f = \int \nu \mathcal{L} d\Omega$. Differentiating $q_B = E_B - |p_B|^2/E_B$ gives $-\Phi(1 + |D_B|^2) + 2D_B \cdot f$, which is (59) because $|\nu| = 1$. $\square$

The finite formula separates remnant staticity, internal directional dispersion of radiation and the relative motion of the two total charges. It extends the additive variance identity without replacing an interacting gravitational energy by a sum of free particle energies: the exact Bondi charges already contain the gravitational dynamics. When $m_B > 0$, (55) also gives $\dot{m}_B = - \int (1 - D_B \cdot \nu) \mathcal{L} d\Omega / S_B \leq 0$. Neither result implies monotonicity of $S_B = m_B/E_B$.

Cone order propagates to the symmetric-power metrics and kernels of Section 7. Their normalized Chern density has no corresponding pointwise order; for example, isotropic loss from a rest charge changes $P = E\mathbf{1}$ while leaving dA_P unchanged. Also, $F_B(n)$ contains only four translation moments. Recovering the angular news, memory or higher BMS charges requires additional boundary data and their charge prescriptions [32]; nonlinear harmonics in the Chern density do not add independent physical information.

9 Quantum staticity and comparison positivity

We restore c and $\hbar$. A positive massive shell supplies useful exact tests, but it is not a construction of the gravitational boundary Hamiltonian. The distinction is especially important when the response observables do not commute.

9.1 A matched massive shell and state dispersion

Fix $Q > 0$ and let $\hat{p}_i$ be momentum multiplication operators on $L^2(\mathbb{R}^3, d^3p; \mathbb{C}^r)$. Define

$$\hat{E} = c\sqrt{Q^2 \mathbf{1} + \hat{p}^2}, \quad \hat{D}_i = c\hat{p}_i \hat{E}^{-1}, \quad \hat{S} = cQ\hat{E}^{-1}, \quad \hat{q} = cQ^2 \hat{E}^{-1}. \quad (60)$$

These are strongly commuting spectral functions. They obey $\sum_i \hat{D}_i^2 + \hat{S}^2 = \mathbf{1}$ and $c\hat{q} = \hat{E}\hat{S}^2$.

Proposition 9.1 (Matched-shell bound). *For $|n| = 1$, on the finite-energy quadratic-form domain,*

$$\hat{F}_n := \hat{E} - cn \cdot \hat{p} = \frac{c}{2}\hat{q} + \frac{\hat{E}}{2} \sum_i (\hat{D}_i - n_i \mathbf{1})^2 \geq \frac{c}{2}\hat{q}. \quad (61)$$

The coefficient $1/2$ is optimal uniformly over momenta and directions.

Proof. The joint spectral functions satisfy $\sum_i (D_i - n_i)^2 = 2 - 2n \cdot D - S^2$. Multiplication by $E/2$ proves the identity. For $p = tn$, the ratio $F_n/(cq)$ tends to $1/2$ as $t/Q \rightarrow \infty$. $\square$

Let ω be a normalized state with finite $e = \omega(\hat{E})$, and let μ_ω be its momentum spectral probability measure. Put $d\nu_E = E d\mu_\omega/e$, $d = c\omega(\hat{p})/e$ and $V_E = \int |D - d|^2 d\nu_E$. The staticity of the mean translation charge, $q_{\text{mean}} = (e^2 - c^2|\omega(\hat{p})|^2)/(ce)$, differs from mean staticity:

$$c(q_{\text{mean}} - \omega(\hat{q})) = eV_E, \quad \omega(\hat{F}_n) = \frac{c}{2}\omega(\hat{q}) + \frac{e}{2}(V_E + |d - n|^2). \quad (62)$$

This follows by expanding the energy-weighted variance. It is valid for pure and mixed states, but uses commuting spectral probabilities and does not isolate a coherence effect. Only the first energy moment is needed since D is bounded. Concavity of $\sqrt{x^2 - c^2Q^2}$ also gives the sharp fixed-mean-energy infimum

$$\omega(\hat{F}_n) \geq e - \sqrt{e^2 - c^2Q^2}. \quad (63)$$

Indeed $c\omega(n \cdot p) \leq c\omega(|p|) \leq \sqrt{e^2 - c^2Q^2}$ by Jensen. Narrow forward momentum packets approach equality; there is no positive constant bound uniform over arbitrarily large e .

9.2 Collective staticity and an attractive internal scale

For additive commuting particle translations, let $E_T = \sum_\alpha E_\alpha$, $p_T = \sum_\alpha p_\alpha$, $D_T = cp_T/E_T$, $Q_T^2 = (E_T/c)^2 - |p_T|^2$ and $q_T = cQ_T^2/E_T$. On the nonvacuum sector,

$$c \left(q_T - \sum_\alpha q_\alpha \right) = \sum_\alpha E_\alpha |D_\alpha - D_T|^2 \geq 0. \quad (64)$$

The proof is again weighted variance using $cq_\alpha = E_\alpha(1 - |D_\alpha|^2)$. Strong joint functional calculus makes this an operator identity in the free theory, including momentum-entangled states. It also explains the algebra of the finite Bondi budget (58); the physical charge balance in that case is supplied by the interacting field equations rather than a free sum.

A controlled binding example is the two-body internal operator

$$\hat{H}_{\text{int}} = 2c\sqrt{\hat{p}^2 + Q^2} - \frac{GQ^2}{c^2r}, \quad Q = m_{\text{phys}}c, \quad \alpha_G = \frac{Gm_{\text{phys}}^2}{\hbar c}. \quad (65)$$

The Kato–Herbst form bound $r^{-1} \leq \pi|p|/(2\hbar)$ gives, for $0 \leq \alpha_G < 4/\pi$,

$$\hat{H}_{\text{int}} \geq 2cQ\sqrt{1 - (\pi\alpha_G/4)^2} \mathbf{1} > 0. \quad (66)$$

To see the coefficient, minimize $\sqrt{t^2 + Q^2} - bt$ for $t \geq 0$, with $b = \pi\alpha_G/4 < 1$; its minimum is $Q\sqrt{1 - b^2}$. Relative form bound below one gives a closed semibounded form. This stability estimate is prior semirelativistic spectral theory [33, 34].

If $\hat{Q}_{\text{int}} = \hat{H}_{\text{int}}/c$ strongly commutes with the external translation momentum, define $\hat{E}_T = c\sqrt{\hat{Q}_{\text{int}}^2 + \hat{p}_T^2}$. Equation (61) then applies with the positive internal scale. This is an internal-mass construction of the familiar Bakamjian–Thomas type [35]. It is a model with an externally chosen G , not a quantum Einstein system or a determination of α_G .

9.3 A bound without commuting momentum components

There is a finite-dimensional extension of the inequality, even though the original sum-of-squares identity need not survive. Let $p_i = p_i^\dagger$ act on a finite-dimensional Hilbert space and define $R = \sum_i p_i^2$, $E = c\sqrt{Q^2\mathbf{1} + R}$ and $q = cQ^2E^{-1}$.

Proposition 9.2 (Noncommuting kinetic-shell bound). *For every unit n , $E - cn \cdot p \geq (c/2)q$. The same optimal uniform coefficient $1/2$ applies.*

Proof. Set $A_n = n \cdot p$. The identity $R - A_n^2 = \sum_i (p_i - n_i A_n)^2 \geq 0$ does not require commuting components. Monotonicity of the operator square root gives $|A_n| \leq \sqrt{R}$, hence

$$E - cA_n \geq c(\sqrt{Q^2\mathbf{1} + R} - \sqrt{R}) = \frac{cQ^2}{\sqrt{Q^2\mathbf{1} + R} + \sqrt{R}} \geq \frac{c}{2}q.$$

The scalar aligned example proves optimality. The square-root step is established operator-monotonicity theory [36, 37]. $\square$

Here cp_iE^{-1} need not be self-adjoint. No commuting MS sphere algebra is inferred. Unbounded versions additionally require common domains and closure of the operator sums; finite matrices suffice for the following distinction.

9.4 Directional positivity and the full Hermitian block

For bounded finite-dimensional responses with $E > 0$, define

$$B = E \otimes \mathbf{1} - c \sum_i p_i \otimes \sigma_i, \quad X_i = cE^{-1/2}p_iE^{-1/2}. \quad (67)$$

Let $\Phi : M_2(\mathbb{C}) \rightarrow \mathcal{B}(\mathcal{H})$ be the unital linear map $\Phi(\mathbf{1}) = \mathbf{1}$, $\Phi(\sigma_i) = X_i$.

Proposition 9.3 (Quantum cone distinction). *All directional inequalities $E - cn \cdot p \geq 0$ are equivalent to positivity of Φ . Positivity of B is equivalent to complete positivity of Φ , and is generally stronger.*

Proof. Rank-one positive matrices are $(\mathbf{1} + n \cdot \sigma)/2$, up to scale. Their images are positive exactly when $\mathbf{1} + n \cdot X \geq 0$ for all n , equivalent by congruence to the directional conditions. The normalized singlet projector is $\Pi_s = (\mathbf{1} \otimes \mathbf{1} - \sum_i \sigma_i \otimes \sigma_i)/4$, so

$$(E^{-1/2} \otimes \mathbf{1})B(E^{-1/2} \otimes \mathbf{1}) = 4(\Phi \otimes \text{id})(\Pi_s).$$

The singlet is related to a maximally entangled projector by a unitary on the second factor. Choi's theorem gives the second equivalence [38]. $\square$

For an explicit shell-compatible counterexample, set $c = 1$, $p_i = -\tau_i/2$, $Q = 1/2$, with Pauli matrices τ_i on the physical two-dimensional space. Then $R = 3\mathbf{1}/4$ and $E = \mathbf{1}$. Every $F_n = \mathbf{1} + n \cdot \tau/2$ is bounded below by $1/2$, whereas $B = \mathbf{1} + \frac{1}{2} \sum_i \tau_i \otimes \sigma_i$ has eigenvalue $-1/2$ on

the singlet. Thus even the massive-shell directional bound does not imply full block positivity. Classical spinor probes test product vectors; they miss this joint-vector obstruction. For strongly commuting translation generators the joint spectral calculus restores the classical equivalence, so the counterexample concerns noncommuting comparison responses rather than ordinary Poincaré translations.

A commutator estimate supplies a sufficient block condition. Put $A = \sum_i p_i \otimes \sigma_i$. Pauli multiplication gives

$$A^2 = R \otimes \mathbf{1} + \mathcal{C}, \quad \mathcal{C} = \frac{i}{2} \sum_{ijk} \epsilon_{ijk} [p_i, p_j] \otimes \sigma_k. \quad (68)$$

If $\mathcal{C} \leq b\mathbf{1}$ with $0 \leq b < Q^2$, square-root monotonicity yields

$$B \geq c(\sqrt{Q^2\mathbf{1} + R} - \sqrt{b\mathbf{1} + R}) \otimes \mathbf{1} \geq \left(1 - \frac{b}{Q^2}\right) \frac{c}{2} q \otimes \mathbf{1}. \quad (69)$$

The endpoint $b = Q^2$ still gives $B \geq 0$. The condition is sufficient, not necessary: for $p_i = a\tau_i$ and $a > 0$, B is positive for every $Q > 0$, although the largest eigenvalue of $\mathcal{C}$ is $6a^2$. An MS application would have to obtain its own commutator bound from the response algebra and constraints.

9.5 Joint response kernels and factorization

A coherent response completion can be formulated without selecting an arbitrary positive ordering. Let $\widehat{K}(x, y)$ be a bounded operator-valued Hermitian kernel on comparison labels. Assume joint Gram positivity,

$$\sum_{a,b} \langle \psi_a, \widehat{K}(x_a, x_b) \psi_b \rangle \geq 0 \quad (70)$$

for every finite collection of labels and state vectors. The operator-valued Kolmogorov–Aronszajn factorization then gives a response Hilbert space and maps $V(x)$ such that $\widehat{K}(x, y) = V(x)^\dagger V(y)$ [39, 40]. One constructs the space from formal pairs (x, ψ) , quotients by the null vectors of (70), and completes. The factorization follows from the induced inner product and is unique up to unitary equivalence in its minimal form.

For a sesquilinear two-spinor kernel with operator coefficients B_{AB} , joint positivity is exactly positivity of the block $[B_{AB}]$ on $\mathcal{H} \otimes \mathbb{C}^2$. Diagonal tests with a single fixed state vector only recover directional positivity. The joint condition is therefore a definite additional target for the MS comparison algebra. Its positivity must be independently justified, and its factorization space is not automatically the physical state space. For an unbounded gravitational boundary form, closability and the identification with the constraint-reduced Hamiltonian remain to be established.

9.6 Covariance, ordering and local energy

Three further tests constrain that identification. First, if a self-adjoint B_0 obeys $U(s)B_0U(s)^{-1} = B_0 + s\beta\mathbf{1}$ for every real s and nonzero β , its spectrum is invariant under all real translations and equals $\mathbb{R}$. A semibounded physical energy therefore cannot inherit this full additive Weyl-shift covariance. Affine positive-variable representations are one established alternative [41].

Second, positive classical squares are not preserved by arbitrary ordering. For $[\widehat{x}, \widehat{p}] = i\hbar\mathbf{1}$ and $\widehat{B}_0 = (\widehat{x}\widehat{p} + \widehat{p}\widehat{x})/2$, the Weyl product gives

$$\text{Op}_W((xp)^2) = \widehat{B}_0^2 - \frac{\hbar^2}{4}\mathbf{1}. \quad (71)$$

The dilation generator $\hat{B}_0$ has spectrum $\mathbb{R}$ in the Schrödinger representation, so the right side has negative spectrum. Choosing an $A^\dagger A$ ordering proves positivity of that chosen operator, but still requires proof that it is the correct physical boundary generator.

Third, local quantum energy densities need not satisfy pointwise DEC even with a positive vacuum-calibrated global Hamiltonian. Quantum energy inequalities control suitable averages [42]; the matter matrix in the classical current is not a universal positive quantum local operator.

At the representation level, suppose $(\hat{E}, c\hat{p})$ are strongly commuting self-adjoint generators with exactly Lorentz-covariant joint spectrum and a finite lower bound for $\hat{E}$. Spacelike or nonzero past-directed spectral points would have Lorentz orbits with arbitrarily negative energy. Hence the spectrum lies in the closed future cone and $\hat{E} - cn \cdot \hat{p} \geq 0$ by functional calculus. This is the standard spectrum-condition argument [19]. For MS, the physical Hilbert space, the constrained generators, their covariance and their stability still have to be constructed. Existing quantum positive-energy proposals, including those of Thiemann and Smolin, are relevant prior work [43, 44].

10 Discussion

The response form gives a finite-dimensional interface between several constructions. Its rank-one tests are projective null probes; its timelike interior gives observer energy; its determinant is the invariant translation scale. The tensor closure equation identifies exactly the four translation modes among general angular responses. The Chern measure supplies their observer-dependent averaging measure, and the mass can be recovered from the mean and variance of those readings.

For Einstein initial data, the matrix Sen–Witten identity realizes this form as a bulk Gram matrix. Its exterior determinant and its induced boundary line curvature are two consequences of the same solution operator. The frame identities exhibit the sign cancellations locally, while the polynomial spinor current keeps the construction regular at spinor zeros. The elliptic lift remains part of the established positive-energy mechanism. A restricted trace-coefficient test does not derive its full phase-sensitive equation from scalar null data.

The Bondi budget gives a dynamical application of the same translation algebra. Radiation is a positive sum of null channels; its invariant determinant measures directional independence. The finite q decomposition separates the remnant, the radiation dispersion and their relative motion. Its differential form has a negative-square loss rate. These statements use the standard gravitational flux law and a fixed observer calibration, and they do not provide the radiative field from its four integrated moments.

The higher-spin construction is a compatible representation lift of the response metric. It connects Chern powers, coherent kernels and global holomorphic norms on the shared projective boundary. Wigner transport, helicity Berry transport and the spatial Sen connection nevertheless remain distinct connections. Their common spinor geometry justifies a precise interface with the companion spin paper, without identifying their physical state spaces or curvature limits.

The quantum calculations show both what survives and what must be added. A chosen positive kinetic shell retains a staticity bound even with noncommuting components. The full operator-valued cone requires joint comparison positivity beyond classical directional probes. Kernel factorization then becomes available, but it cannot establish the missing physical positivity assumption or the boundary Hamiltonian identification. Weyl shifts, ordering and local negative energy place further restrictions on that construction.

Finally, none of these normalized boundary structures determines the radial gravitational scale. The Chern measure is invariant under a positive rescaling of the charge; the Sen factorization contains κ and the Einstein constraints as inputs; the Bondi news flux contains G in its normalization. The dimensionless attractive coupling α_G is likewise prescribed in the model. A derivation of Newton’s constant would require an independently fixed microscopic response

action and a controlled matching to these asymptotic generators. The present results specify that matching problem more precisely, while keeping the established kinematics and gravitational theorems at their appropriate level.

References

- [1] Mingyang Han, From Local Response to Global Gravitational Charges: Asymptotic Momentum and Stationary Boundaries in Motion–Staticity Geometry (2026). Unpublished companion manuscript.
- [2] Richard Arnowitt, Stanley Deser and Charles W. Misner, The Dynamics of General Relativity, in *Gravitation: An Introduction to Current Research*, edited by Louis Witten (Wiley, 1962), pp. 227–265. [arXiv:gr-qc/0405109](https://arxiv.org/abs/gr-qc/0405109).
- [3] Tullio Regge and Claudio Teitelboim, Role of Surface Integrals in the Hamiltonian Formulation of General Relativity, *Annals of Physics* **88**, 286–318 (1974). [doi:10.1016/0003-4916\(74\)90404-7](https://doi.org/10.1016/0003-4916(74)90404-7).
- [4] Mingyang Han, Spin at the Closed Motion–Staticity Boundary: Rest-State Topology, Wigner Reduction, and Little-Group Contraction (2026). Unpublished companion manuscript.
- [5] Roger Penrose and Wolfgang Rindler, *Spinors and Space-Time: Two-Spinor Calculus and Relativistic Fields*, vol. 1 (Cambridge University Press, Cambridge, 1984). [doi:10.1017/CB09780511564048](https://doi.org/10.1017/CB09780511564048).
- [6] Richard Schoen and Shing-Tung Yau, On the Proof of the Positive Mass Conjecture in General Relativity, *Communications in Mathematical Physics* **65**, 45–76 (1979). [doi:10.1007/BF01940959](https://doi.org/10.1007/BF01940959).
- [7] Edward Witten, A New Proof of the Positive Energy Theorem, *Communications in Mathematical Physics* **80**, 381–402 (1981). [doi:10.1007/BF01208277](https://doi.org/10.1007/BF01208277).
- [8] Thomas Parker and Clifford H. Taubes, On Witten’s Proof of the Positive Energy Theorem, *Communications in Mathematical Physics* **84**, 223–238 (1982). [doi:10.1007/BF01208569](https://doi.org/10.1007/BF01208569).
- [9] James M. Nester, Special Orthonormal Frames, *Journal of Mathematical Physics* **33**, 910–913 (1992). [doi:10.1063/1.529742](https://doi.org/10.1063/1.529742).
- [10] James M. Nester and Roh-Suan Tung, Another Positivity Proof and Gravitational Energy Localizations, *Physical Review D* **49**, 3958–3962 (1994). [doi:10.1103/PhysRevD.49.3958](https://doi.org/10.1103/PhysRevD.49.3958).
- [11] Thomas H. Parker, Gauge Choice in Witten’s Energy Expression, *Communications in Mathematical Physics* **100**, 471–480 (1985). [doi:10.1007/BF01217725](https://doi.org/10.1007/BF01217725).
- [12] Miguel Campiglia, Null to time-like infinity Green’s functions for asymptotic symmetries in Minkowski spacetime, *Journal of High Energy Physics* **2015** (11), 160 (2015). [doi:10.1007/JHEP11\(2015\)160](https://doi.org/10.1007/JHEP11(2015)160).
- [13] Eric Palmerduca and Hong Qin, Helicity Is a Topological Invariant of Massless Particles: $C = -2h$, *Physical Review Research* **7**, L022001 (2025). [doi:10.1103/PhysRevResearch.7.L022001](https://doi.org/10.1103/PhysRevResearch.7.L022001).
- [14] Jean-Pierre Serre, Représentations linéaires et espaces homogènes kählériens des groupes de Lie compacts, in *Séminaire Bourbaki: années 1951/52–1952/53–1953/54, exposés 50–100* (Société mathématique de France, 1954), pp. 447–454. https://www.numdam.org/item/SB_1951-1954__2__447_0/.

- [15] A. M. Perelomov, Coherent States for Arbitrary Lie Group, *Communications in Mathematical Physics* **26** (3), 222–236 (1972). doi:[10.1007/BF01645091](https://doi.org/10.1007/BF01645091).
- [16] Hermann Bondi, M. G. J. van der Burg and A. W. K. Metzner, Gravitational Waves in General Relativity. VII. Waves from Axi-Symmetric Isolated Systems, *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences* **269** (1336), 21–52 (1962). doi:[10.1098/rspa.1962.0161](https://doi.org/10.1098/rspa.1962.0161).
- [17] R. K. Sachs, Gravitational Waves in General Relativity. VIII. Waves in Asymptotically Flat Space-Time, *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences* **270** (1340), 103–126 (1962). doi:[10.1098/rspa.1962.0206](https://doi.org/10.1098/rspa.1962.0206).
- [18] Thomas Mädler and Jeffrey Winicour, Bondi–Sachs Formalism, *Scholarpedia* **11** (12), 33528 (2016). doi:[10.4249/scholarpedia.33528](https://doi.org/10.4249/scholarpedia.33528).
- [19] Eugene P. Wigner, On Unitary Representations of the Inhomogeneous Lorentz Group, *Annals of Mathematics* **40** (1), 149–204 (1939). doi:[10.2307/1968551](https://doi.org/10.2307/1968551).
- [20] Yongdo Lim, Hilbert’s projective metric on Lorenz cones and Birkhoff formula for Lorentzian compressions, *Linear Algebra and its Applications* **423** (2–3), 246–254 (2007). doi:[10.1016/j.laa.2006.12.015](https://doi.org/10.1016/j.laa.2006.12.015).
- [21] Jeffrey M. Bowen and James W. York, Jr., Time-Asymmetric Initial Data for Black Holes and Black-Hole Collisions, *Physical Review D* **21**, 2047–2056 (1980). doi:[10.1103/PhysRevD.21.2047](https://doi.org/10.1103/PhysRevD.21.2047).
- [22] Steven Brandt and Bernd Brügmann, A Simple Construction of Initial Data for Multiple Black Holes, *Physical Review Letters* **78**, 3606–3609 (1997). doi:[10.1103/PhysRevLett.78.3606](https://doi.org/10.1103/PhysRevLett.78.3606).
- [23] Jörg Frauendiener, James M. Nester and László B. Szabados, Witten Spinors on Maximal, Conformally Flat Hypersurfaces, *Classical and Quantum Gravity* **28** (18), 185004 (2011). doi:[10.1088/0264-9381/28/18/185004](https://doi.org/10.1088/0264-9381/28/18/185004).
- [24] Sven Hirsch, Demetre Kazaras and Marcus Khuri, Spacetime Harmonic Functions and the Mass of 3-Dimensional Asymptotically Flat Initial Data for the Einstein Equations, *Journal of Differential Geometry* **122**, 223–258 (2022). arXiv:[2002.01534](https://arxiv.org/abs/2002.01534).
- [25] Hubert L. Bray, Demetre Kazaras, Marcus A. Khuri and Daniel L. Stern, Harmonic Functions and the Mass of 3-Dimensional Asymptotically Flat Riemannian Manifolds, *Journal of Geometric Analysis* **32**, 184 (2022). doi:[10.1007/s12220-022-00915-9](https://doi.org/10.1007/s12220-022-00915-9).
- [26] Robert Bartnik, The Mass of an Asymptotically Flat Manifold, *Communications on Pure and Applied Mathematics* **39**, 661–693 (1986). doi:[10.1002/cpa.3160390505](https://doi.org/10.1002/cpa.3160390505).
- [27] Amitabha Sen, Gravity as a Spin System, *Physics Letters B* **119** (1–3), 89–91 (1982). doi:[10.1016/0370-2693\(82\)90250-7](https://doi.org/10.1016/0370-2693(82)90250-7).
- [28] J. M. Radcliffe, Some Properties of Coherent Spin States, *Journal of Physics A: General Physics* **4** (3), 313–323 (1971). doi:[10.1088/0305-4470/4/3/009](https://doi.org/10.1088/0305-4470/4/3/009).
- [29] Norbert Dragon, Bundle Structure of Massless Unitary Representations of the Poincare Group, *International Journal of Theoretical Physics* **63**, 149 (2024). doi:[10.1007/s10773-024-05612-z](https://doi.org/10.1007/s10773-024-05612-z).
- [30] Blagoje Oblak, Probing Wigner Rotations for Any Group, *Journal of Geometry and Physics* **129**, 168–185 (2018). doi:[10.1016/j.geomphys.2018.03.008](https://doi.org/10.1016/j.geomphys.2018.03.008).

- [31] Gabriele Veneziano and Gregory A. Vilkovisky, Angular Momentum Loss in Gravitational Scattering, Radiation Reaction, and the Bondi Gauge Ambiguity, *Physics Letters B* **834**, 137419 (2022). [doi:10.1016/j.physletb.2022.137419](https://doi.org/10.1016/j.physletb.2022.137419).
- [32] Robert M. Wald and Andreas Zoupas, General Definition of “Conserved Quantities” in General Relativity and Other Theories of Gravity, *Physical Review D* **61** (8), 084027 (2000). [doi:10.1103/PhysRevD.61.084027](https://doi.org/10.1103/PhysRevD.61.084027).
- [33] Ira W. Herbst, Spectral Theory of the Operator $(p^2 + m^2)^{1/2} - Ze^2/r$, *Communications in Mathematical Physics* **53**, 285–294 (1977). [doi:10.1007/BF01609852](https://doi.org/10.1007/BF01609852).
- [34] Richard L. Hall and Wolfgang Lucha, Gravitating Semirelativistic N -Boson Systems, *Journal of Physics A: Mathematical and General* **39**, 11531–11540 (2006). [doi:10.1088/0305-4470/39/37/013](https://doi.org/10.1088/0305-4470/39/37/013).
- [35] Benjamin Bakamjian and Lawrence H. Thomas, Relativistic Particle Dynamics. II, *Physical Review* **92**, 1300–1310 (1953). [doi:10.1103/PhysRev.92.1300](https://doi.org/10.1103/PhysRev.92.1300).
- [36] Karl Löwner, Über monotone Matrixfunktionen, *Mathematische Zeitschrift* **38**, 177–216 (1934). [doi:10.1007/BF01170633](https://doi.org/10.1007/BF01170633).
- [37] Erhard Heinz, Beiträge zur Störungstheorie der Spektralzerlegung, *Mathematische Annalen* **123**, 415–438 (1951). [doi:10.1007/BF02054965](https://doi.org/10.1007/BF02054965).
- [38] Man-Duen Choi, Completely Positive Linear Maps on Complex Matrices, *Linear Algebra and its Applications* **10** (3), 285–290 (1975). [doi:10.1016/0024-3795\(75\)90075-0](https://doi.org/10.1016/0024-3795(75)90075-0).
- [39] Nachman Aronszajn, Theory of Reproducing Kernels, *Transactions of the American Mathematical Society* **68** (3), 337–404 (1950). [doi:10.1090/S0002-9947-1950-0051437-7](https://doi.org/10.1090/S0002-9947-1950-0051437-7).
- [40] Jan Stochel, Decomposition and Disintegration of Positive Definite Kernels on Convex *-Semigroups, *Annales Polonici Mathematici* **56** (3), 243–294 (1992). <https://eudml.org/doc/262273>.
- [41] John R. Klauder, Noncanonical Quantization of Gravity. I. Foundations of Affine Quantum Gravity, *Journal of Mathematical Physics* **40**, 5860–5882 (1999). [doi:10.1063/1.533052](https://doi.org/10.1063/1.533052).
- [42] Christopher J. Fewster and Simon P. Eveson, Bounds on Negative Energy Densities in Flat Spacetime, *Physical Review D* **58**, 084010 (1998). [doi:10.1103/PhysRevD.58.084010](https://doi.org/10.1103/PhysRevD.58.084010).
- [43] Thomas Thiemann, QSD VI: Quantum Poincaré Algebra and a Quantum Positivity of Energy Theorem for Canonical Quantum Gravity, *Classical and Quantum Gravity* **15**, 1463–1485 (1998). [doi:10.1088/0264-9381/15/6/006](https://doi.org/10.1088/0264-9381/15/6/006).
- [44] Lee Smolin, Positive Energy in Quantum Gravity, *Physical Review D* **90**, 044034 (2014). [doi:10.1103/PhysRevD.90.044034](https://doi.org/10.1103/PhysRevD.90.044034).